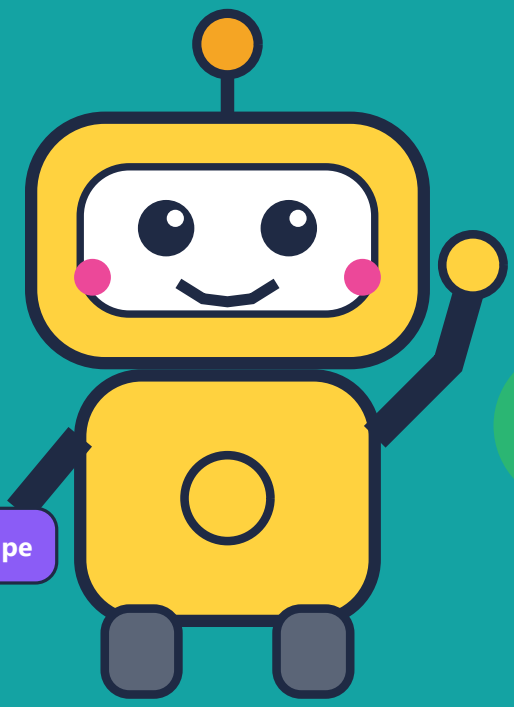
CodeSchool · Class 6

Think Like a Coder

Computational thinking and AI for Class 6

Learn it · See it · Try it · Build it



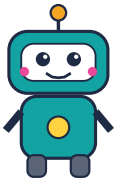
Follows NCERT Class 6 maths, CBSE's four AI chapters and two projects

Built for CBSE's Computational Thinking and AI curriculum, 2026-27

Theory, worked examples, practice, coding on paper, AI and projects

SAMPLE EDITION · DRAFT

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Hello again! I am Robo. This year we go further: we will write functions, work with lists and data, find out how AI learns from examples, and run two real projects. All you need is a pencil and a curious mind.

How this book works

Learn	Robo explains one big idea in simple words, with the new coder words to remember.
Example	A worked example shows the idea step by step.
Try it!	Five practice questions. Circle the answer or write it in the box.
Now you build it	Coding on paper: put code cards in order, fix bugs and write your own programs.
Challenge	A harder puzzle for when you are ready for more.
How did I do?	Colour a face to show how sure you feel.
Projects	Two 20-hour team projects: plan, collect real data, build, test, present and score your work with a rubric.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this **Repeat: a loop** **If: a decision** **Define: a function**

When: an event

For teachers and parents

Chapters 1-10 go with the same chapters of the NCERT Class 6 maths textbook, so they can be used in maths periods. AI chapters 1-4 follow the AI part of CBSE's Class 6 handbook, and the two projects run across the year. CBSE sets 100 hours a year in Class 6: about 40 for computational thinking, 20 for AI and 40 for two projects. The coding here needs no computer: students write, trace and debug programs on paper, which prepares them for block and text coding. Answers are at the back. Encourage children to explain their thinking, not just the answer.

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Coder skills in this book

Levels: C1 rules, functions and lists; C2 algorithms with inputs, sorting and debugging; C3 AI ideas, data and projects.

Ch	Coder skill	Level	Ch	Coder skill	Level
1	Rules and functions: name a rule and reuse it	C1	9	Reflect and rotate: functions that change shapes	C2
2	Drawing programs: angles, turns and a square(size) function	C1	10	Integers in data: lowest, highest and range	C2
3	Algorithms that repeat until something happens	C1	AI1	Tell AI from automation; how machines learn	C3
4	Lists: store, sort and find the biggest	C1	AI2	Types of data; collect, organise and show it	C3
5	The sieve and a factor-finding algorithm	C2	AI3	Learn a rule from labelled examples, then decide	C3
6	Functions with two inputs; breaking shapes apart	C2	AI4	Strong passwords, privacy and digital footprints	C3
7	Compare, simplify and sort fractions	C2	P1	Collect, organise and present real data	C3
8	Construction scripts: exact steps and debugging	C2	P2	Design, test and improve an algorithm	C3

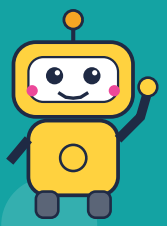
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CHAPTER 1 · LEVEL C1

1 Patterns in Mathematics

Coder skill: Rules and functions: name a rule and reuse it

Goes with NCERT maths, Class 6, chapter 1

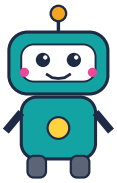


Learn

A **sequence** is a list of numbers that follows a **rule**. Some rules say how to get the **next** term (add 3 each time). Others say how to get the term at any **position** (term = 3 x position + 2), which lets you jump straight to the 100th term.

Coders give a useful rule a name and call it a **function**. A function takes an **input** and gives back an **output**: *square(4)* gives back 16, *triangle(4)* gives back 1 + 2 + 3 + 4 = 10.

sequence numbers that follow a rule	**term** one number in a sequence	**function** a named rule with an input	**output** what the function gives back



A rule I use again and again gets a name: a function. square(n) means n x n. Give it 5 and it gives back 25!

Example

Adding odd numbers makes squares

1. 1 = 1, 1 + 3 = 4, 1 + 3 + 5 = 9, 1 + 3 + 5 + 7 = 16. The answers are **square numbers**.
2. The picture shows why: each new odd number is an L-shaped layer that turns a square into the next square.
3. So the first **n** odd numbers add up to **n x n**. The first 10 odd numbers add up to 100.



★ *Coder fact!* Spreadsheet formulas work like functions: write the rule once and it fills thousands of rows.

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Try it!

Circle the right answer, or write it in the box.

1 Write the next two terms: 1, 4, 9, 16, 25, ...

2 Triangular numbers: 1, 3, 6, 10, 15, ... What comes next?

3 A sequence starts at 5 and adds 4 each time. What is its 10th term?

a) 41 **b)** 45 **c)** 40 **d)** 37

4 The function **f(n) = 3 x n - 1**. What is f(4)? Which input n gives the output 29?

5 In 2, 3, 3, 4, 4, 4, 5, 5, 5, 5, ... each number appears one time fewer than its value. How many times does 7 appear?

a) 5 **b)** 6 **c)** 7 **d)** 8

Now you build it

Write a function called **triangle(n)** that adds 1 + 2 + ... + n and gives back the total.

Define triangle(n) **Set total to 0** **Repeat for k = 1 to n** **change total by k** **give back total**

Run triangle(4) and triangle(6) by hand. What does each give back?

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★ Challenge

Find the sum of the first 20 odd numbers without adding them one by one. Explain your shortcut.

✓ How did I do?

I can describe a sequence with a rule.	I can use a function with an input.	I can spot square and triangular numbers.
		

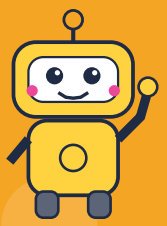
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CHAPTER 2 · LEVEL C1

2 Lines and Angles

Coder skill: Drawing programs: angles, turns and a square(size) function

Goes with NCERT maths, Class 6, chapter 2

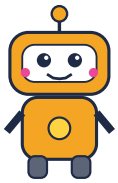


Learn

An **angle** is an amount of turn, measured in **degrees**. A right angle is 90°, a straight angle 180° and a full turn 360°. Angles less than 90° are **acute**; between 90° and 180° are **obtuse**; more than 180° are **reflex**.

A drawing robot uses *forward* and *turn*. To draw a closed shape with equal sides, it turns 360° ÷ number of sides. Put the steps in a **function** with an input, like *square(size)*, and reuse it.

angle	acute	obtuse	reflex
an amount of turn	less than 90°	between 90° and 180°	more than 180°

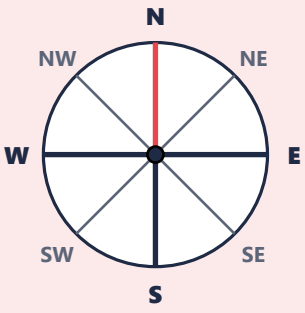


I draw with two commands: forward and turn. Wrap them in a function called square(size) and I can draw any size of square!

Example

Turning on the 8-point compass

1. Neighbouring directions on an 8-point compass are **45°** apart (360° ÷ 8).
2. Face North and turn 135° clockwise: 135 ÷ 45 = 3 steps, through NE and E, to **SE**.
3. Face West and turn 90° anticlockwise: 2 steps, through SW, to **South**.



★ *Coder fact!* Ships and aircraft steer by headings in degrees: 90 means due East and 180 means due South.

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Try it!

Circle the right answer, or write it in the box.

1 What type of angle is this?



a) acute **b)** right **c)** obtuse **d)** reflex

2 You face North-East and turn 90° anticlockwise. Which way do you face now?

a) North-West **b)** South-East **c)** North **d)** East

3 What angle do the hands of a clock make at 4 o'clock?

4 Fill the gap to draw a regular hexagon: *repeat 6 times: forward 40, turn right* ___ °

5 Eight children stand evenly around a circle, facing the centre. What angle at the centre is between two neighbours?

a) 30° **b)** 45° **c)** 60° **d)** 90°

Now you build it

Write a **square(size)** function and use it twice.

Define square(size) **Repeat 4 times** **forward size** **turn right 90°**

square(20) **square(40)**

Now change it into **triangle(size)** for a triangle with equal sides. What goes in the repeat and the turn?

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★ Challenge

Using square(30) and turns only, write a program that draws a windmill of 4 squares around one point.

✓ How did I do?

I can name and measure angles.	I can turn on an 8-point compass.	I can write and reuse a drawing function.
		

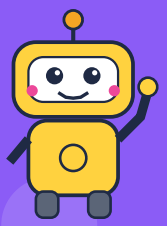
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CHAPTER 3 · LEVEL C1

3 Number Play

Coder skill: Algorithms that repeat until something happens

Goes with NCERT maths, Class 6, chapter 3



Learn

An **algorithm** is a set of exact steps. Some algorithms **repeat until** a condition is true, so you cannot tell in advance how many steps they take.

Number puzzles are full of them. *Reverse and add until you get a palindrome* (a number that reads the same both ways, like 363). To check an algorithm, **trace** it: write down every value as it changes.

algorithm exact steps to solve a problem	**repeat until** loop that stops when something is true	**palindrome** reads the same both ways	**trace** follow steps and write each value



Some loops don't know in advance how many times to run. They repeat UNTIL something happens, like reaching a palindrome!

Example

Reverse and add, starting with 57

1. 57 is not a palindrome, so add its reverse: 57 + 75 = 132.
2. 132 is not a palindrome: 132 + 231 = 363.
3. 363 is a palindrome, so the loop **stops** after 2 steps.

Step	Number	Reverse and add
1	57	57 + 75 = 132
2	132	132 + 231 = 363
stop	363	palindrome

Coder fact! The Indian mathematician D. R. Kaprekar found that any 4-digit number with at least two different digits reaches 6174 with his routine.

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Try it!

Circle the right answer, or write it in the box.

1 Which of these is a palindrome?

a) 1231 **b)** 4554 **c)** 4545 **d)** 4500

2 Start with 39 and reverse-and-add until you get a palindrome. How many steps does it take, and what palindrome do you reach?

3 In the row 32, 47, 18, 59, 26, a number is a **supercell** if it is bigger than its neighbours on both sides (or its one neighbour, at an end). Which numbers are supercells?

4 Rule: *if the number is even, halve it; if it is odd, multiply by 3 and add 1. Repeat until you reach 1.* Starting at 6, how many steps does it take?

5 In a dice game the score is the total of all rolls, and the game ends when a 6 comes up for the second time. The final score is 20. What is the least number of rolls?

a) 3 **b)** 4 **c)** 5 **d)** 6

Now you build it

Kaprekar's routine for 3-digit numbers (digits not all the same).

Input a 3-digit number

Repeat until the number is 495

make the biggest number from its digits

make the smallest number from its digits

set number to biggest - smallest

change steps by 1

Trace it for 352. Write each subtraction. How many steps does it take?

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★ Challenge

Run Kaprekar's routine on 211. What happens? (Hint: keep three digits, so 99 is written 099.)

✓ How did I do?

I can follow and trace an algorithm.	I can use 'repeat until'.	I can explore number puzzles like palindromes.
		

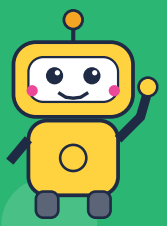
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CHAPTER 4 · LEVEL C1

4 Data Handling and Presentation

Coder skill: Lists: store, sort and find the biggest

Goes with NCERT maths, Class 6, chapter 4

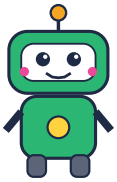


Learn

Data is a collection of facts, such as marks, temperatures or votes. Coders store data in a **list**: items in order, each at a numbered **position**.

Data is easier to understand when organised in a **frequency table** and shown in a **pictograph** or **bar graph**. Every graph needs a title, labels and a **scale**, such as 1 square = 5 students.

data facts collected to answer a question	**list** items in order, by position	**frequency** how often something occurs	**scale** what one unit stands for

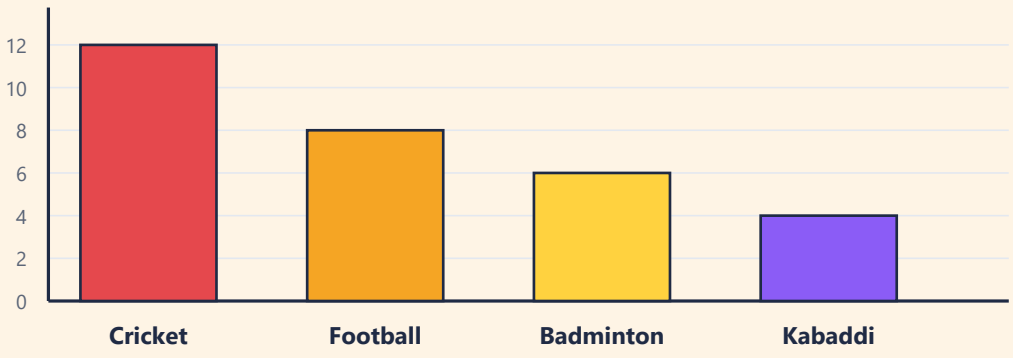


I keep data in a list. Each item has a position: first, second, third... Then I can sort it, search it or draw it!

Example

Favourite sports of 30 students

1. Survey results as a list: cricket 12, football 8, badminton 6, kabaddi 4. Check: 12 + 8 + 6 + 4 = 30.
2. In the bar graph, the height of each bar shows the number of students.
3. Sorted from most to least: cricket, football, badminton, kabaddi. Cricket is 3 times as popular as kabaddi.



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Try it!

Circle the right answer, or write it in the box.

1 Here is a list. What is in position 3? What is the largest value?

	1	2	3	4	5
temps =	31	28	33	30	35

2 Each symbol stands for 4 trees. How many more trees did Class 6 plant than Class 7?



3 In a bar graph, 1 square stands for 5 students. A bar is 7 squares tall. How many students does it show?

4 Sort 42, 17, 58, 23 from smallest to largest. Which number is in position 2?

a) 17 **b)** 23 **c)** 42 **d)** 58

5 In a pictograph each symbol stands for x litres, where x is a whole number. A shop has 6 symbols and sold less than 100 litres. What is the largest x can be?

a) 15 **b)** 16 **c)** 17 **d)** 20

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Now you build it

An algorithm that finds the biggest item in a list.

Set biggest to the first item | Repeat for each item in the list | If item > biggest | set biggest to item | say biggest

Trace it on the list 12, 19, 7, 25, 14. Write the value of biggest after each item.

Challenge

Heights in cm: 132, 140, 128, 145, 137. Change the program so it finds the shortest. Which cards change, and what is the answer?

How did I do?

I can store data in a list.	I can read pictographs and bar graphs with a scale.	I can find the biggest item with an algorithm.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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5

CHAPTER 5 · LEVEL C2

Prime Time

Coder skill: The sieve and a factor-finding algorithm

Goes with NCERT maths, Class 6, chapter 5

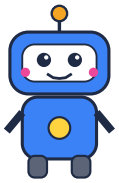


Learn

A **factor** divides a number exactly; a **multiple** is in its times table. A **prime** has exactly two factors, 1 and itself. 1 is not prime.

The **Sieve of Eratosthenes** is a 2,000-year-old algorithm: cross out 1; circle 2 and cross out its other multiples; circle the next number not crossed out (3) and cross out its multiples; repeat. The circled numbers are the primes.

factor divides a number exactly	**multiple** a number in the times table	**prime** exactly two factors	**sieve** an algorithm that filters



A sieve keeps what you want and lets the rest fall through. My number sieve keeps only the primes!

Example

The sieve for 1 to 30

1. Cross out 1. Keep 2, cross out 4, 6, 8, ... Keep 3, cross out 9, 15, 21, 27 (6, 12, ... are already gone).
2. Keep 5 and cross out 25. The next prime, 7, has 7 x 7 = 49, which is past 30, so we can **stop**.
3. The primes up to 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29.

~~1~~	**2**	**3**	~~4~~	**5**	~~6~~	**7**	~~8~~	~~9~~	~~10~~
11	~~12~~	**13**	~~14~~	~~15~~	~~16~~	**17**	~~18~~	**19**	~~20~~
~~21~~	~~22~~	**23**	~~24~~	~~25~~	~~26~~	~~27~~	~~28~~	**29**	~~30~~

★ *Coder fact!* Very large prime numbers help keep online banking and messages secret.

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Try it!

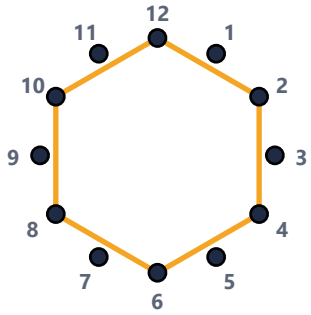
Circle the right answer, or write it in the box.

1 How many prime numbers are there from 1 to 30?

2 Which of these is a prime number?

a) 21 **b)** 27 **c)** 29 **d)** 33

3 12 points are equally spaced on a circle. Start at 2 and join every 2nd point: 2, 4, 6, ... How many lines are drawn before you get back to 2?



a) 3 **b)** 4 **c)** 5 **d)** 6

4 On the same circle, start at 5 and join every 5th point: 5, 10, 3, ... How many lines are drawn before you get back to 5?

a) 4 **b)** 5 **c)** 6 **d)** 12

5 How many factors does 36 have?

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Now you build it

An algorithm that lists the factors of n and checks if n is prime.

Input n | Repeat for k = 1 to n | If n ÷ k leaves no remainder | add k to the list factors | If factors has exactly 2 items | say 'prime'

Run it for n = 15 and n = 13. What is in the list each time, and does it say 'prime'?

Challenge

On the 12-point circle, which jumps from 1 to 11 visit every point? What do these numbers have in common with 12?

How did I do?

I can find factors and multiples.	I can use the sieve to find primes.	I can explain a factor-finding algorithm.
		

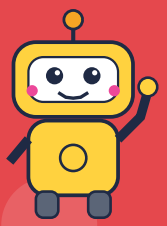
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CHAPTER 6 · LEVEL C2

6 Perimeter and Area

Coder skill: Functions with two inputs; breaking shapes apart

Goes with NCERT maths, Class 6, chapter 6

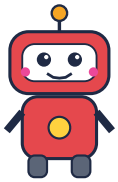


Learn

Perimeter is the distance around a shape; **area** is the space inside it. For a rectangle: perimeter = 2 x (l + w) and area = l x w.

A function can have more than one input: *area(l, w)*. To find the area of an L-shape, **decompose** it into rectangles, find each area and add. This is the same skill coders use to break a big problem into smaller ones.

perimeter	area	decompose	input
the distance around	the space inside	break into smaller parts	a value given to a function

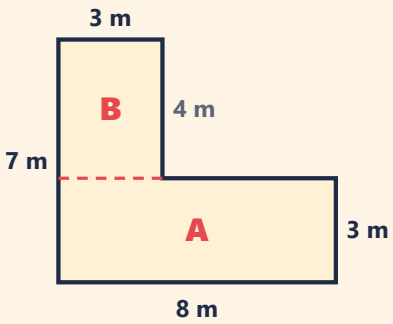


area(l, w) has two inputs. I break a tricky shape into rectangles, call area() for each part and add the answers!

Example

The area and perimeter of an L-shaped garden

1. Split the L into rectangle A (8 m by 3 m) and rectangle B (3 m by 4 m).
2. area(8, 3) = 24 and area(3, 4) = 12, so the garden is 24 + 12 = **36 m²**.
3. Perimeter: add the six sides: 8 + 3 + 5 + 4 + 3 + 7 = **30 m**. (The top step is 8 - 3 = 5 m.)



★ *Coder fact!* Architects' software finds the area of a floor plan by breaking it into rectangles and triangles.

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Try it!

Circle the right answer, or write it in the box.

1 What are area(7, 5) and the perimeter of a 7 cm by 5 cm rectangle?

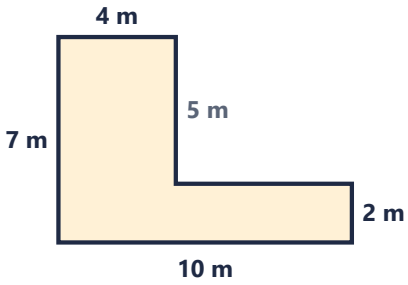
2 A square has a perimeter of 20 cm. What is its area?

a) 20 cm^2 **b)** 25 cm^2 **c)** 16 cm^2 **d)** 100 cm^2

3 Two rectangles both have an area of 24 cm^2: 6 cm by 4 cm, and 12 cm by 2 cm. Which has the longer perimeter?

a) 6 by 4 **b)** 12 by 2 **c)** the same

4 Find the area of this shape by breaking it into two rectangles.



5 A square of perimeter 16 cm is cut along a diagonal into two triangles. Which statement is **false**?

a) Two sides of each triangle meet at a right angle

b) Each triangle has a side of 4 cm

c) Together the triangles have an area of 16 cm^2

d) Each triangle has an area of 16 cm^2

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Now you build it

Use one function inside another.

Define area(l, w)

give back l x w

Define floor_cost(l, w, rate)

give back area(l, w) x rate

say floor_cost(5, 4, 60)

A classroom is 5 m by 4 m and tiles cost Rs 60 per square metre. What does the program say?

Challenge

Draw two shapes, each made of 12 unit squares joined edge to edge: one with the shortest perimeter you can find, one with the longest.

How did I do?

I can find perimeter and area.	I can use a function with two inputs.	I can break a shape into rectangles.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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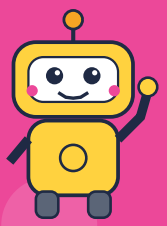
CHAPTER 7 · LEVEL C2

7

Fractions

Coder skill: Compare, simplify and sort fractions

Goes with NCERT maths, Class 6, chapter 7

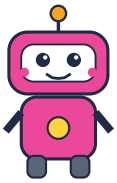


Learn

Equivalent fractions name the same amount: 1/2 = 2/4 = 3/6. A fraction is in **simplest form** when no number above 1 divides both top and bottom.

To **compare** fractions, write them with the same bottom number (or as marks out of 100). To **sort** a list, a computer can compare neighbours and swap them when they are in the wrong order, going along the list again and again (**bubble sort**).

equivalent the same amount	**simplest form** cannot be simplified more	**compare** find which is bigger	**bubble sort** swap neighbours until sorted



To sort fractions, I compare two neighbours and swap them if they are in the wrong order. Then I do it again!

Example

Which subject did Riya do best in?

1. Riya's marks: Art 18/20, Maths 42/50, Science 88/100. Different totals make them hard to compare.
2. Make them all out of 100: 18/20 = 90/100, 42/50 = 84/100, 88/100 stays 88/100.
3. Sorted from best: **Art, Science, Maths**.

Subject	Marks	Out of 100
Art	18/20	90
Maths	42/50	84
Science	88/100	88

Coder fact! Music uses fractions: a half note lasts half as long as a whole note.

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Try it!

Circle the right answer, or write it in the box.

1 What is 18/24 in simplest form?

a) 9/12 **b)** 3/4 **c)** 6/8 **d)** 2/3

2 Which is bigger: 5/8 or 3/5?

a) 5/8 **b)** 3/5 **c)** equal

3 Put 2/3, 3/8 and 1/2 in order from smallest. Which one is in the middle?

a) 2/3 **b)** 3/8 **c)** 1/2

4 One pass of bubble sort on [3/4, 1/4, 1/2]: compare each pair of neighbours from the left and swap if the left one is bigger. What is the list after one pass?

a) [1/4, 1/2, 3/4] **b)** [1/4, 3/4, 1/2] **c)** [3/4, 1/2, 1/4]

5 Asha eats 1/3 of a pizza and Ben eats 1/4. What fraction is left?

Now you build it

An algorithm that puts a fraction in simplest form.

Input top and bottom | **Repeat until no number above 1 divides both** | **find a number that divides both** | **divide top and bottom by it** | **say top / bottom**

Trace it for 24/36. Write each division you make.

Challenge

Put 7/8, 5/6 and 11/12 in order without drawing. What pattern helps?

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✓ How did I do?

I can simplify fractions.	I can compare and order fractions.	I can trace a sorting pass.
		

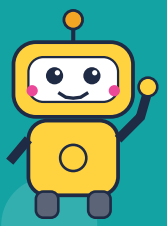
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CHAPTER 8 · LEVEL C2

8 Playing with Constructions

Coder skill: Construction scripts: exact steps and debugging

Goes with NCERT maths, Class 6, chapter 8

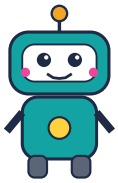


Learn

A **construction** uses only a ruler and a compass, following exact steps. It is an algorithm: anyone who follows the same steps gets the same drawing.

All points the same distance from a centre make a **circle**. Where two circles cross, a point is that distance from *both* centres. If a script gives the wrong shape, **debug** it: follow it step by step and find the step that is missing or wrong.

construct draw exactly with ruler and compass	**arc** part of a circle	**perpendicular** at right angles	**debug** find and fix a mistake

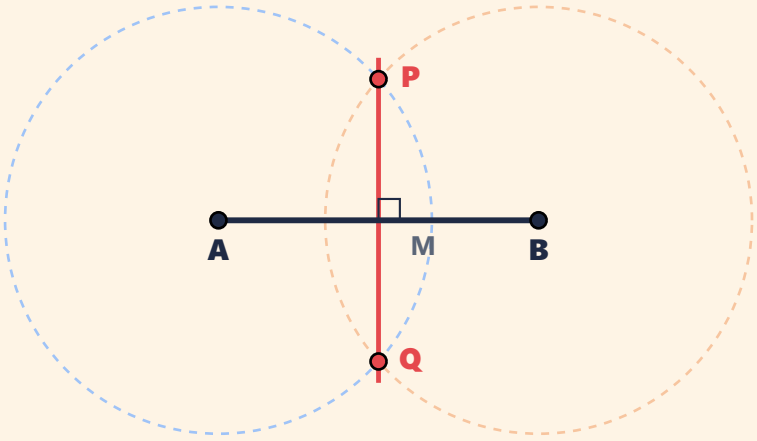


A construction is a program for your compass and ruler. If one step is missing, the drawing has a bug!

Example

A script for the perpendicular bisector of AB

1. Draw AB = 6 cm.
2. With the compass at A and a radius more than 3 cm, draw arcs above and below AB.
3. With the same radius and the compass at B, draw arcs that cross the first ones at P and Q.
4. Join P and Q. PQ cuts AB at its middle point M, at a right angle.



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Try it!

Circle the right answer, or write it in the box.

1 Two circles of radius 5 cm have centres 6 cm apart. At how many points do they cross?

a) 0 **b)** 1 **c)** 2 **d)** 3

2 Two circles of radius 3 cm have centres 8 cm apart. At how many points do they cross?

a) 0 **b)** 1 **c)** 2 **d)** 3

3 Debug this script for a 4 cm square: *Draw AB = 4 cm. At B draw a right angle and mark BC = 4 cm. Join A to C.* What goes wrong?

a) It draws a triangle, not a square **b)** Nothing is wrong

c) The sides are too long

4 All the points that are exactly 3 cm from a point O make a ...

a) line **b)** circle **c)** square **d)** triangle

5 A 2 by 2 grid is made of 4 small squares. How many squares can you see in it altogether?

Now you build it

Follow this script with a ruler and set square to draw a rectangle 5 cm by 3 cm.

Draw AB = 5 cm | **At A and B, draw right angles** | **Mark AD = 3 cm and BC = 3 cm** | **Join D to C** | **Check: ___**

Draw it in the space. Then write a check step that would catch a mistake.

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★ Challenge

Write a script that uses only a ruler and compass to draw a triangle with all three sides 5 cm.

✓ How did I do?

I can write exact construction steps.	I can use circles to find points.	I can debug a script that goes wrong.
		

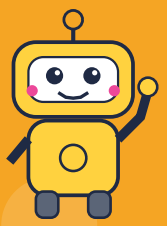
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CHAPTER 9 · LEVEL C2

9 Symmetry

Coder skill: Reflect and rotate: functions that change shapes

Goes with NCERT maths, Class 6, chapter 9

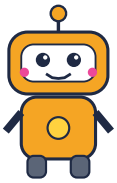


Learn

A shape has a **line of symmetry** if folding along the line makes both halves match. It has **rotational symmetry** if it looks the same after less than a full turn; the number of times it matches in one full turn is its **order**.

Drawing programs flip shapes with a **reflect** function. On a grid with a mirror line, each square moves to the same distance on the other side of the line.

line of symmetry	reflection	rotational symmetry	order
fold line where halves match	a mirror image	matches itself when turned	how many times it matches

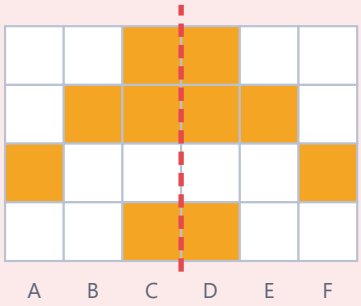


reflect() is a function too: give it a square on one side of the mirror, and it gives back the square on the other side!

Example

Reflecting a shape on a grid

1. The red line is the mirror. A square 1 column left of the line goes to 1 column right of it.
2. A square 3 columns left goes 3 columns right. The row does not change.
3. Reflect every shaded square and the whole shape is mirrored.



Coder fact! Photo apps flip pictures with a reflect function: every pixel moves to the other side of a mirror line.

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Try it!

Circle the right answer, or write it in the box.

1 How many lines of symmetry does a square have?

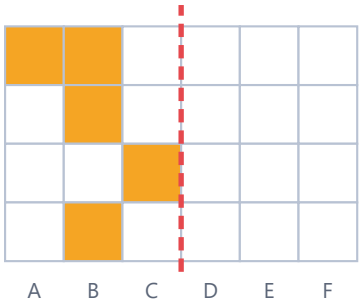
2 Which letter has **no** line of symmetry?

a) A **b)** H **c)** F **d)** O

3 What is the order of rotational symmetry of an equilateral triangle?

a) 1 **b)** 2 **c)** 3 **d)** 6

4 Shade the mirror image on the other side of the red line. How many squares will be shaded in all?



5 A square sheet is folded in half. A triangle is cut out with one side along the fold. What do you see when it is unfolded?

a) a triangle-shaped hole **b)** a diamond-shaped hole

c) two separate triangle holes **d)** no hole

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Now you build it

Columns on a grid are numbered 1 to 6, and the mirror line is between columns 3 and 4.

Define reflect(column)

give back 7 - column

Repeat for each shaded square

shade column reflect(column) in the same row

A shape covers columns 1, 2 and 3 of a row. Which columns does its reflection cover? Check that reflect(3) is 4.

Challenge

Design a pattern with exactly 2 lines of symmetry and rotational symmetry of order 2. Draw it and mark the lines.

How did I do?

I can find lines of symmetry.	I can find the order of rotational symmetry.	I can use a reflect function on a grid.
		

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10

CHAPTER 10 · LEVEL C2

The Other Side of Zero

Coder skill: Integers in data: lowest, highest and range

Goes with NCERT maths, Class 6, chapter 10

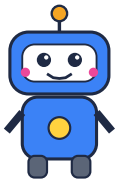


Learn

Integers are the whole numbers with their negatives: ..., -3, -2, -1, 0, 1, 2, 3, ... On a number line, numbers to the left are smaller, so -7 is less than -3.

Integers appear in real data: winter temperatures in Leh, floors below the ground in a lift, or a bank balance. The **range** of data is highest minus lowest; with negatives, subtracting a negative means adding: 4 - (-8) = 12.

integer	negative	range	minimum
a whole number or its negative	less than zero	highest minus lowest	the lowest value

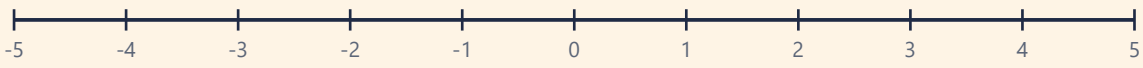


Below zero is not 'nothing'. -8 °C is colder than -3 °C. I track the lowest and highest values, even when they are negative!

Example

Tracing a lift

1. The lift starts at the ground floor, 0. It goes up 4 floors: 0 + 4 = **4**.
2. It goes down 6 floors: 4 - 6 = **-2** (the second basement).
3. It goes up 1 floor: -2 + 1 = **-1**.



Coder fact! Lifts in tall buildings show basement floors as B1 and B2, or as -1 and -2.

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Try it!

Circle the right answer, or write it in the box.

1 Which is smaller?

a) -7 **b)** -3

2 At night it is 3 °C. The temperature falls by 8 degrees. What is it now?

3 P50, Q42, R34, S26, ... Which letter is the first to have a negative number?

a) U **b)** V **c)** W **d)** X

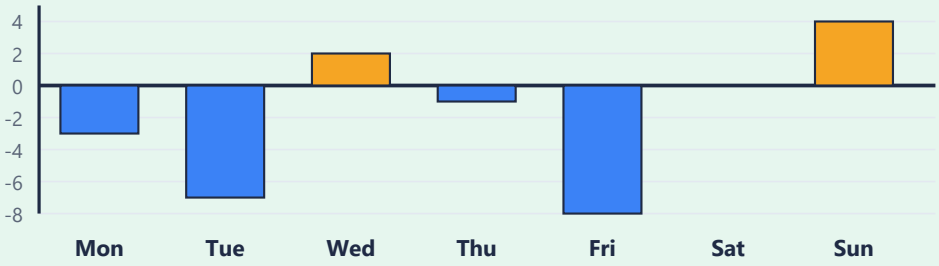
4 A lift starts at floor 2. It goes down 5, up 1, then down 3. Which floor is it on?

5 Work out (-4) + 9 + (-6).

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Now you build it

A temperature log for one week in Leh in winter (°C).



Set lowest and highest to the first value

Repeat for each temperature

If temp < lowest, set lowest to temp

If temp > highest, set highest to temp

say highest - lowest

Run it on the week: -3, -7, 2, -1, -8, 0, 4. What are the lowest, the highest and the range?

Challenge

A submarine is at -120 m. It rises 45 m, dives 30 m, then rises 60 m. Where is it now? Above or below sea level?

How did I do?

I can compare and order integers.	I can add and subtract with negatives.	I can find the range of data with negatives.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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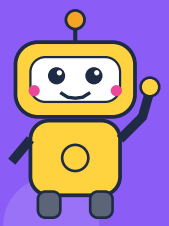
AI1

AI CHAPTER 1 · LEVEL C3

Introduction to AI

AI skill: Tell AI from automation; how machines learn

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 1



Learn

Intelligence is the ability to learn, think, solve problems and adapt. **Artificial Intelligence (AI)** makes machines do tasks that usually need human intelligence, such as recognising faces or understanding speech, by finding patterns in **data**.

Automation follows fixed instructions and does the same thing every time, like a washing machine timer. AI **learns**: from labelled examples (**supervised** learning), by finding groups on its own (**unsupervised**), or by trial, reward and penalty (**reinforcement**).

AI machines that learn and decide	**automation** fixed steps, same result	**labelled data** examples with the right answer	**model** what an AI learns



A timer that always does the same thing is automation. A machine that learns from examples and gets better is AI!

Example

AI or automation?

1. A traffic light that changes every 60 seconds follows a fixed rule: **automation**.
2. Face unlock on a phone learnt your face from pictures and still works with glasses on: **AI**.
3. Ask: does it **learn from data** and improve, or always do the same thing?

Machine	Learns from data?	AI or automation
Microwave timer	No	Automation
Spam filter	Yes	AI
Voice assistant	Yes	AI
Fixed-time signal	No	Automation

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Try it!

Circle the right answer, or write it in the box.

1 Which of these is **not** AI?

a) a voice assistant
b) face recognition
c) a fixed-time traffic signal
d) a chatbot

2 A teacher gives a computer 500 photos, each tagged 'cat' or 'dog'. This is ...

a) labelled data
b) unlabelled data
c) not data

3 A game program gets +1 for a win and -1 for a loss, and plays thousands of games to get better. This is ...

a) supervised learning
b) unsupervised learning
c) reinforcement learning

4 A shop's computer puts customers into groups by what they buy, without being told what the groups are. This is ...

a) supervised learning
b) unsupervised learning
c) reinforcement learning

5 The first research project to use the name 'Artificial Intelligence' was held in ...

a) 1906
b) 1956
c) 1996
d) 2016

Now you build it

Write a sorting rule for 'AI or automation?' as code cards.

Look at the machine | **If it learns from data or examples** | **put it in the AI box** | **else** | **put it in the automation box**

Use your rule on: a calculator, a spam filter, an alarm clock, a photo app that finds faces, a music app that suggests songs, street lights on a timer.

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★ Challenge

Pick a machine at home that follows fixed instructions. How could AI make it more helpful? What data would it need?

✓ How did I do?

I can explain what AI is.	I can tell AI from automation.	I can name three ways machines learn.
		

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AI2

AI CHAPTER 2 · LEVEL C3

Basic Data Concepts

AI skill: Types of data; collect, organise and show it

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 2

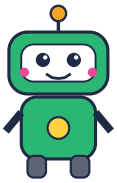


Learn

Data is raw facts: numbers, words, pictures, sounds or videos. When it is organised to answer a question it becomes **information**.

We **collect** data by asking (surveys), watching (observation), measuring, or from records. We **organise** it by grouping similar items (classifying), putting it in tables, labelling it clearly and storing it safely. We **show** it with tables, bar charts and pictograms.

data raw facts	**information** data organised to be useful	**survey** asking people questions	**classify** group similar items

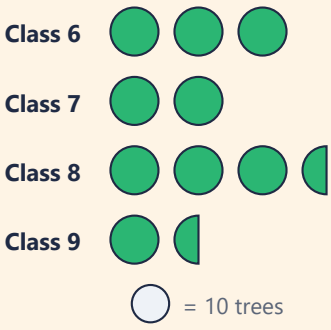


AI is only as good as its data. So first we learn to collect it carefully, organise it neatly and show it clearly!

Example

The eco club's tree-planting drive

1. Data collected: Class 6 planted 30 trees, Class 7 20, Class 8 35 and Class 9 15.
2. Key: one tree symbol = 10 trees, so 35 trees is 3 and a half symbols.
3. Information: Class 8 planted the most, 20 more than Class 9.



★ *Coder fact!* Weather apps combine data from satellites, radars and thousands of weather stations.

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Try it!

Circle the right answer, or write it in the box.

1 A recording of the school bell is which type of data?

a) number **b)** text **c)** image **d)** sound

2 Measuring a plant's height every week collects which type of data?

a) number **b)** text **c)** image **d)** sound

3 In a pictogram, one symbol = 10 trees. Class 8 has 3½ symbols and Class 9 has 1½. How many more trees did Class 8 plant?

4 Putting similar data into groups is called ...

a) labelling **b)** classifying **c)** storing **d)** deleting

5 A teacher wants to compare five students' marks at a glance. What is best?

a) a bar chart **b)** a long paragraph

c) a sound recording

Now you build it

Plan and run a small survey: *How do you come to school?*

Ask one clear question **Collect answers with tally marks**

Organise into a frequency table **Draw a bar chart with a scale** **Write one conclusion**

Ask 10 classmates. Draw your tally table and bar chart in the space, then write your conclusion.

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★ Challenge

A survey asked 5 players from the cricket team, 'Which sport is best?' All said cricket. Can we say the whole school likes cricket best? Why?

✓ How did I do?

I can name types of data.	I can collect and organise data.	I can choose a table or chart to show data.
		

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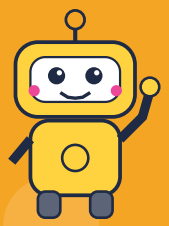
AI3

AI CHAPTER 3 · LEVEL C3

Patterns and Decisions

AI skill: Learn a rule from labelled examples, then decide

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 3

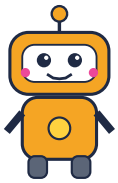


Learn

A **pattern** is something that repeats or can be predicted. AI finds patterns in **labelled examples** and uses them to **predict** labels for new things. This is how a spam filter learns which emails are spam.

An **observation** says what the data shows (it was warmer at 2 pm on 5 days). A **conclusion** says what it means (afternoons are usually warmer). A conclusion from a little data may be wrong, so we test it, and use more and fairer data.

pattern	observation	conclusion	prediction
something that repeats	what the data shows	what the data means	a best guess for a new case



I learn like this: look at examples with the right answers, find the pattern, make a rule, then test it on new examples!

Example

Learning when a mango is ripe

1. Look at the labelled examples. Colour does not decide it: mango 4 is yellow but not ripe, and mango 5 is green but ripe.
2. Every ripe mango is soft and every unripe one is hard. Rule: **if soft, then ripe**.
3. Test it on new mangoes whose labels we know. If it is right on most of them, we can use it to decide.

Mango	Colour and feel	Label
1	yellow, soft	ripe
2	green, hard	not ripe
3	yellow, soft	ripe
4	yellow, hard	not ripe
5	green, soft	ripe

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Try it!

Circle the right answer, or write it in the box.

1 'On 5 days it was warmer at 2 pm than at 8 am.' This is ...

a) an observation **b)** a conclusion

2 'So afternoons are usually warmer than mornings.' This is ...

a) an observation **b)** a conclusion

3 Labelled emails: *Win a prize now!* spam; *Class test on Monday* not spam; *Free gift, click here* spam; *PTM at 10 am* not spam. What would a filter trained on these predict for *Click now to win free coins*?

a) spam **b)** not spam

4 An AI learnt 'cat' only from photos of white cats. It is shown a black cat. What might go wrong?

a) It may not recognise it as a cat **b)** It will always be right

c) It will turn the photo white

5 Riya scored 18, 19 and 20 in weeks when she practised daily, and 12 and 11 in weeks when she did not. What is the best decision?

a) practise daily **b)** stop practising

c) nothing can be decided

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Now you build it

Be the classifier. Find the rule that tells a **Zib** from a **Zob**.

Training examples (labelled)



New creatures to classify



Look at the new creature | If it has ___ | predict Zib | else | predict Zob

Fill in the rule, then label the three new creatures. Does colour matter?

Challenge

A new creature has both stripes and spots. What should the classifier say? How could we make it better?

How did I do?

I can tell an observation from a conclusion.	I can find a rule from labelled examples.	I can explain why more and fairer data helps.
☺ 😐 ☹	☺ 😐 ☹	☺ 😐 ☹

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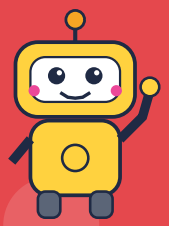
AI4

AI CHAPTER 4 · LEVEL C3

Ethics and Digital Responsibility

AI skill: Strong passwords, privacy and digital footprints

Goes with the AI part of CBSE's Class 6 CT&AI handbook, chapter 4

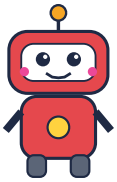


Learn

Digital responsibility means being careful, respectful and honest when using technology. Copying someone's work as your own is **plagiarism**; copying paid software without permission is **piracy**; tricking people into giving passwords is **phishing**.

Everything you do online leaves a **digital footprint**. **Active** footprints are things you choose to share, like posts and forms. **Passive** ones are collected without you noticing, like location data and cookies. Protect your **privacy**, use strong passwords, and tell a trusted adult if something feels wrong.

digital footprint the trail you leave online	**phishing** tricking people to get secrets	**plagiarism** copying work as your own	**privacy** control of your personal data



Online, think before you click, share or copy. Be safe, be kind and be honest, just like in the classroom!

Example

Make a strong password you can remember

1. Pick three random words that do not belong together: *mango, kite, river*.
2. Join them with a symbol and add a number: **Mango!Kite7River**. It is long and mixes capitals, small letters, a number and a symbol.
3. Never use your name or birthday, never reuse it everywhere, and never share it, not even with friends.

strong

★ *Coder fact!* In India, cyber fraud can be reported on the helpline 1930 or at cybercrime.gov.in.

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Try it!

Circle the right answer, or write it in the box.

1 Which password is strongest?

a) riya2014
b) password
c) Kite!Mango7River
d) 123456

2 An email says: *Your account is locked. Reply with your password to unlock it.* This is ...

a) phishing
b) a prize
c) a software update

3 Posting a photo of your science project online leaves which kind of footprint?

a) active
b) passive

4 An app records where you are without you posting anything. This is which kind of footprint?

a) active
b) passive

5 Arjun copies an essay from a website and hands it in as his own. This is ...

a) plagiarism
b) research
c) hacking
d) piracy

Now you build it

A safety rule for messages, written as code cards.

When a message asks for personal information
If you do not know the sender
do not reply or click; tell an adult
else
check with an adult before sharing

Write your own rule for this: a stranger in a game chat asks for the name of your school.

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★ Challenge

List 5 things that are part of your digital footprint. Mark each one active or passive.

✓ How did I do?

I can make and protect a strong password.	I can spot phishing.	I can explain active and passive footprints.
		

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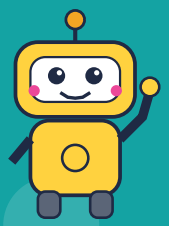
P1

PROJECT 1 · 20 HOURS · LEVEL C3

Weather Across India

Project skill: Collect, organise and present real data

CBSE: 40 project hours a year in Class 6, as two projects of 20 hours



The big question

How different is the weather in five Indian cities, and what patterns can we find?

Your team picks five cities in different parts of India, for example Leh, Shimla, Jaipur, Kolkata and Chennai. For two weeks, record each city's highest and lowest temperature and whether it rained, from the newspaper or an official source such as the India Meteorological Department (IMD).

Then organise the data, find the highest, lowest and range for each city, draw charts, look for patterns and present what you found. This project uses lists, integers, min and max, charts, patterns and conclusions.

dataset	source	range	trend
all the data you collected	where the data came from	highest minus lowest	a change over time

✓ My project plan

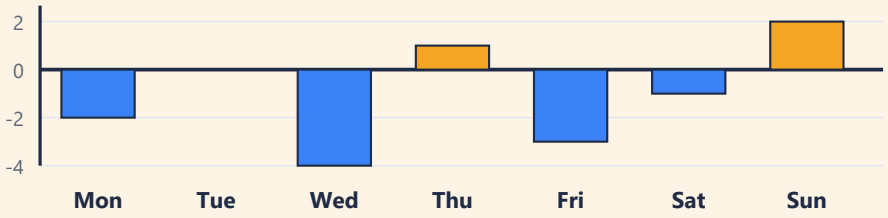
Weeks	Stage	What I do	Done
1	**Define**	Agree the question, choose five cities and decide what to record each day.	☐
2-3	**Collect**	Record the highest and lowest temperature and rain for each city every day for two weeks. Write the source.	☐
4	**Organise**	Put the data in one table per city. Look for missing or odd values and check them.	☐
5	**Calculate**	For each city find the highest, the lowest and the range. Count the rainy days.	☐
6	**Show**	Draw a bar chart of the ranges, and a bar chart of one city's daily highs.	☐
7	**Find patterns**	Write three observations and one conclusion. Is two weeks of data enough to be sure?	☐
8	**Predict and test**	Predict tomorrow's highest temperature for one city, then check the real value.	☐
9	**Present**	Make a poster or a 3-minute talk: question, data, charts, findings and limits.	☐
10	**Reflect**	Score your work with the rubric and write what you would improve.	☐

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How a coder does it

From raw data to a finding: Leh, one week in January (practice data)

1. Daily highest temperatures: -2, 0, -4, 1, -3, -1, 2 °C.
2. Highest 2, lowest -4, so the range is 2 - (-4) = **6 degrees**.
3. Observation: the highest temperature was below 0 °C on 4 of the 7 days.
4. Conclusion, with its limit: Leh's January days are often below freezing, but one week of data is not enough to be sure.



My data sheet

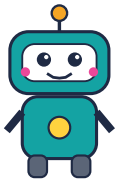
One row per day for one city. Write where each value came from.

Date	City	High °C	Low °C	Rain? (Y/N)	Source

Draw your chart

Draw a bar chart of the temperature range of each city. Choose a scale, then label both axes and give the chart a title.

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A project is like a big program: plan it, break it into parts, test it and fix it. One stage at a time!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Question	Clear, and can be answered with the data collected.	1 2 3 4	1 2 3 4
Data	Complete, from a named source, with odd values checked.	1 2 3 4	1 2 3 4
Thinking	Highest, lowest and range are right; patterns are spotted.	1 2 3 4	1 2 3 4
Charts and talk	Charts have titles, scales and labels; the talk is clear.	1 2 3 4	1 2 3 4
Honesty and limits	Says what the data cannot show; nothing copied without credit.	1 2 3 4	1 2 3 4
Teamwork	Everyone has a job and the work is shared fairly.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] I wrote down where every piece of data came from.
- [ ] I did not change any data to make a better story.
- [ ] I said what my data cannot tell us (only two weeks, only five cities).
- [ ] I used only official or trusted websites, with an adult's help.

Looking back

What surprised you most in your data?

If you did this project again, what would you change?

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P2

PROJECT 2 · 20 HOURS · LEVEL C3

A Smarter School Routine

Project skill: Design, test and improve an algorithm

CBSE: 40 project hours a year in Class 6, as two projects of 20 hours



The big question

Can we design a better routine for a real problem at school, and prove it works?

Pick a real problem at school: a long queue at the water cooler, books returned to the wrong shelf, or a slow start to assembly. Observe it and collect data, such as how many minutes the queue takes.

Design a new routine as an algorithm and draw it as a flowchart with a start, steps, decisions and an end. Try it for a week, collect the same data again and compare. Did your algorithm make things better?

flowchart a picture of an algorithm	**decision** a yes or no question in a flowchart	**test** try it and measure	**improve** change it to work better

My project plan

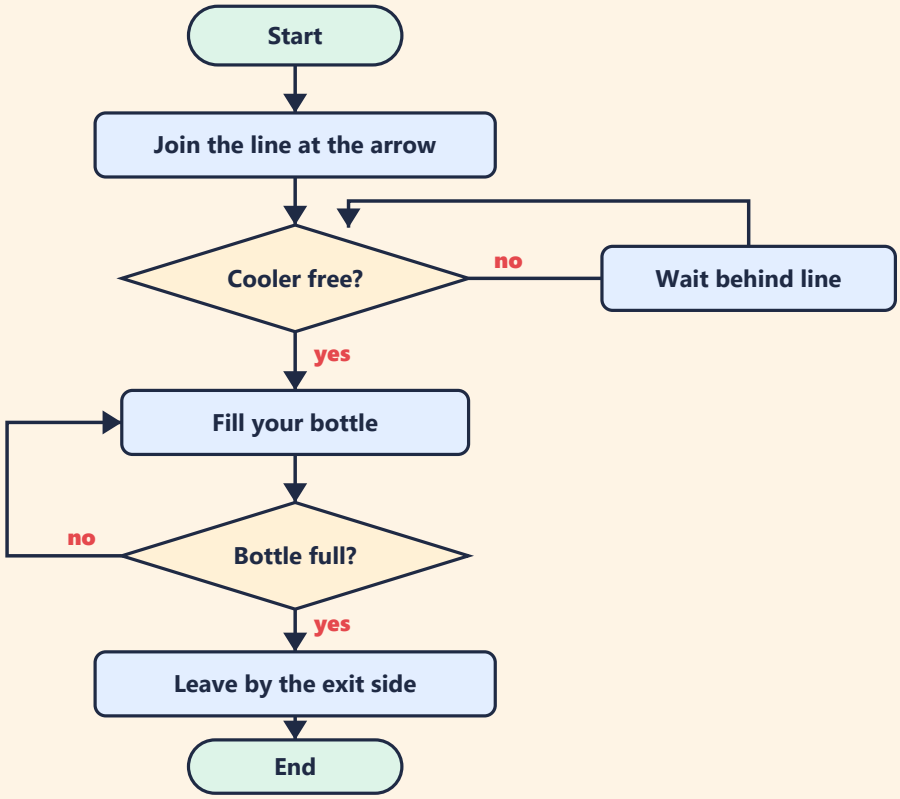
Weeks	Stage	What I do	Done
1	**Define**	Choose one problem. Write who it affects and how you will measure it.	☐
2	**Observe**	Collect 'before' data: for example, time five lunch breaks at the water cooler.	☐
3-4	**Design**	Break the problem down. Write your new routine as steps, then draw it as a flowchart.	☐
5	**Check**	Trace the flowchart with a friend. Fix any step that is unclear or missing.	☐
6-7	**Try it**	With your teacher's permission, use the routine for a week and collect 'after' data.	☐
8	**Compare**	Put the before and after data in a table and a bar chart.	☐
9	**Present**	Show the problem, the flowchart, the data and the result to your class.	☐
10	**Reflect**	Score your work with the rubric. Was the routine fair to everyone?	☐

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How a coder does it

A flowchart for the water cooler

1. Ovals show **start** and **end**, boxes show **steps** and diamonds show **decisions**, with a 'yes' and a 'no' arrow.
2. The 'no' arrows make loops: wait until the cooler is free; keep filling until the bottle is full.
3. Test result (practice data): 20 children took 9 minutes before and 6 minutes after, which is 3 minutes faster.



Before and after data

Measure the same thing in the same way, on five days before and five days after the new routine.

Day	What we measured	Before	After	Difference
Example	Minutes for 20 children to fill bottles	9	6	3

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Draw your flowchart

Use ovals for start and end, boxes for steps and diamonds for decisions. Label both arrows from every diamond. Trace it with a friend before you try it for real.

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Coders don't guess. They measure, change one thing, and measure again!

★ How good is my project?

Score each row from 1 (just starting) to 4 (excellent). Then your teacher scores it too. Talk about any row where the scores differ.

What we check	What excellent (4) looks like	Me	Teacher
Problem	A real, specific problem with a clear way to measure it.	1 2 3 4	1 2 3 4
Algorithm	Steps are exact, in order, and cover what to do when something goes wrong.	1 2 3 4	1 2 3 4
Flowchart	Correct shapes; every decision has a 'yes' and a 'no' path.	1 2 3 4	1 2 3 4
Testing	Before and after data are collected the same way and compared.	1 2 3 4	1 2 3 4
Fairness	The routine works for everyone, including younger children.	1 2 3 4	1 2 3 4
Talk and teamwork	A clear talk with the flowchart and chart; the work is shared fairly.	1 2 3 4	1 2 3 4

✓ Fair, safe and honest?

- [ ] We asked permission before trying the routine.
- [ ] We collected no names or personal details, only counts and times.
- [ ] We listened to people who did not like the new routine.
- [ ] We reported the real result, even if things did not improve.

Looking back

Which step of your algorithm needed the most fixing, and why?

Who else in school could use your routine?

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Answers

Many questions have more than one good answer. Accept any answer that follows the rule and can be explained.

1. Patterns in Mathematics

1. 36 and 49
2. 21 (add 6)
3. a) 41 (5 + 9 x 4)
4. f(4) = 11; n = 10
5. b) 6
Build it: triangle(4) = 10; triangle(6) = 21
Challenge: 400: the first n odd numbers add up to n x n, and 20 x 20 = 400

2. Lines and Angles

1. c) obtuse (120°)
2. a) North-West
3. 120° (4 x 30°)
4. 60
5. b) 45°
Build it: Repeat 3 times: forward size, turn right 120° (3 x 120° = 360°)
Challenge: Repeat 4 times: square(30), turn right 90° (each square starts at the centre)

3. Number Play

1. b) 4554
2. 2 steps: 39 + 93 = 132, 132 + 231 = 363
3. 47 and 59
4. 8 steps: 6, 3, 10, 5, 16, 8, 4, 2, 1
5. b) 4: two 6s make 12; the other 8 needs at least 2 rolls with no 6 (such as 4 + 4)
Build it: 532 - 235 = 297, 972 - 279 = 693, 963 - 369 = 594, 954 - 459 = 495: 4 steps
Challenge: 211 - 112 = 099, 990 - 099 = 891, 981 - 189 = 792, 972 - 279 = 693, 963 - 369 = 594, 954 - 459 = 495: it still reaches 495, in 6 steps

4. Data Handling and Presentation

1. 33 and 35
2. 6 (16 - 10)
3. 35
4. b) 23
5. b) 16 (6 x 16 = 96; 6 x 17 = 102 is too many)
Build it: 12, 19, 19, 25, 25: it says 25
Challenge: Use 'smallest' instead of 'biggest' and 'If item < smallest'; the shortest is 128 cm

5. Prime Time

1. 10
2. c) 29
3. d) 6
4. d) 12 (5 and 12 share no factor except 1, so every point is visited)
5. 9: 1, 2, 3, 4, 6, 9, 12, 18, 36
Build it: 15: [1, 3, 5, 15], not prime; 13: [1, 13], prime
Challenge: 1, 5, 7 and 11: they share no factor with 12 except 1

6. Perimeter and Area

1. 35 cm^2 and 24 cm
2. b) 25 cm^2 (side 5 cm)
3. b) 12 by 2 (28 cm, against 20 cm)
4. 40 m^2 (10 x 2 + 4 x 5)
5. d) each triangle has an area of 8 cm^2, not 16
Build it: area(5, 4) = 20; 20 x 60 = Rs 1,200
Challenge: Shortest: a 3 x 4 rectangle (14 units). Longest: a 1 x 12 strip, or any shape whose squares never form a 2 x 2 block (26 units)

7. Fractions

1. b) 3/4
2. a) 5/8 (25/40 against 24/40)
3. c) 1/2 (3/8, 1/2, 2/3)
4. a) [1/4, 1/2, 3/4]
5. 5/12 (1 - 4/12 - 3/12)
Build it: For example ÷2: 12/18, ÷2: 6/9, ÷3: 2/3 (or ÷12 in one step)
Challenge: 5/6, 7/8, 11/12: each is 1 minus a piece (1/6, 1/8, 1/12); the smaller the missing piece, the bigger the fraction

8. Playing with Constructions

1. c) 2
2. a) 0 (3 + 3 is less than 8)
3. a) it draws a triangle; it needs: at C draw a right angle, mark CD = 4 cm, join D to A
4. b) circle
5. 5 (4 small and 1 big)
Build it: For example: check DC = 5 cm and that the two diagonals are equal (about 5.8 cm)
Challenge: Draw AB = 5 cm; draw an arc of radius 5 cm with centre A; draw an arc of radius 5 cm with centre B; they cross at C; join AC and BC

9. Symmetry

1. 4
2. c) F
3. c) 3
4. 10
5. b) a diamond-shaped hole
Build it: Columns 6, 5 and 4 (7 - 1, 7 - 2 and 7 - 3)
Challenge: Open answer, for example a rectangle or the letter H

10. The Other Side of Zero

1. a) -7
2. -5 °C
3. c) W (T18, U10, V2, W-6)
4. -5
5. -1
Build it: Lowest -8, highest 4, range 4 - (-8) = 12 degrees
Challenge: -45 m (-120 + 45 - 30 + 60), which is 45 m below sea level

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AI 1. Introduction to AI

1. c) a fixed-time traffic signal
2. a) labelled data
3. c) reinforcement learning
4. b) unsupervised learning
5. b) 1956, at Dartmouth College
Build it: AI: spam filter, face-finding photo app, song suggestions. Automation: calculator, alarm clock, timer street lights
Challenge: Open answer, for example a fan that learns when the family is usually home, using data about times and temperatures

AI 2. Basic Data Concepts

1. d) sound
2. a) number
3. 20 (35 - 15)
4. b) classifying
5. a) a bar chart
Build it: Open answer; check that the tallies add up to 10 and that the chart has a title, labels and a scale
Challenge: No: 5 people is a small sample, and they all play cricket, so the data is biased

AI 3. Patterns and Decisions

1. a) an observation
2. b) a conclusion
3. a) spam (it has the words 'win', 'free' and 'click')
4. a) it may not recognise it: its training data was not varied enough
5. a) practise daily
Build it: If it has stripes, predict Zib, else Zob. New creatures: Zib, Zob, Zib. Colour does not matter: each colour appears in both groups
Challenge: It cannot be sure, because it never saw one like this; it should say 'not sure', and we should add labelled examples like it to the training data

AI 4. Ethics and Digital Responsibility

1. c) Kite!Mango7River
2. a) phishing
3. a) active
4. b) passive
5. a) plagiarism
Build it: For example: if a stranger asks where I live or study, I do not answer, I block or report them, and I tell a trusted adult
Challenge: Open answer, for example active: posts, comments, online forms; passive: location history, cookies, suggestions based on your likes

Project 1. Weather Across India

Open project: assess with the rubric. Check that each range is highest minus lowest, for example 2 - (-4) = 6.

Project 2. A Smarter School Routine

Open project: assess with the rubric. A good flowchart has one start, one end, and a 'yes' and a 'no' arrow from every decision.

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Coder words

acute	less than 90°	Ch 2
AI	machines that learn and decide	AI 1
algorithm	exact steps to solve a problem	Ch 3
angle	an amount of turn	Ch 2
arc	part of a circle	Ch 8
area	the space inside	Ch 6
automation	fixed steps, same result	AI 1
bubble sort	swap neighbours until sorted	Ch 7
classify	group similar items	AI 2
compare	find which is bigger	Ch 7
conclusion	what the data means	AI 3
construct	draw exactly with ruler and compass	Ch 8
data	facts collected to answer a question	Ch 4
dataset	all the data you collected	Project 1
debug	find and fix a mistake	Ch 8
decision	a yes or no question in a flowchart	Project 2
decompose	break into smaller parts	Ch 6
digital footprint	the trail you leave online	AI 4
equivalent	the same amount	Ch 7
factor	divides a number exactly	Ch 5
flowchart	a picture of an algorithm	Project 2
frequency	how often something occurs	Ch 4
function	a named rule with an input	Ch 1
improve	change it to work better	Project 2
information	data organised to be useful	AI 2
input	a value given to a function	Ch 6
integer	a whole number or its negative	Ch 10
labelled data	examples with the right answer	AI 1
line of symmetry	fold line where halves match	Ch 9
list	items in order, by position	Ch 4
minimum	the lowest value	Ch 10
model	what an AI learns	AI 1
multiple	a number in the times table	Ch 5
negative	less than zero	Ch 10
observation	what the data shows	AI 3
obtuse	between 90° and 180°	Ch 2
order	how many times it matches	Ch 9

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output	what the function gives back	Ch 1
palindrome	reads the same both ways	Ch 3
pattern	something that repeats	AI 3
perimeter	the distance around	Ch 6
perpendicular	at right angles	Ch 8
phishing	tricking people to get secrets	AI 4
plagiarism	copying work as your own	AI 4
prediction	a best guess for a new case	AI 3
prime	exactly two factors	Ch 5
privacy	control of your personal data	AI 4
range	highest minus lowest	Ch 10
reflection	a mirror image	Ch 9
reflex	more than 180°	Ch 2
repeat until	loop that stops when something is true	Ch 3
rotational symmetry	matches itself when turned	Ch 9
scale	what one unit stands for	Ch 4
sequence	numbers that follow a rule	Ch 1
sieve	an algorithm that filters	Ch 5
simplest form	cannot be simplified more	Ch 7
source	where the data came from	Project 1
survey	asking people questions	AI 2
term	one number in a sequence	Ch 1
test	try it and measure	Project 2
trace	follow steps and write each value	Ch 3
trend	a change over time	Project 1

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Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

Define ___	give back ___	Repeat ___ times
Define ___	give back ___	Repeat ___ times
Repeat for each item	Repeat until ___	If ___
Repeat for each item	Repeat until ___	If ___
else	When ___ happens	Input ___
else	When ___ happens	Input ___
Set ___ to ___	change ___ by ___	say ___
Set ___ to ___	change ___ by ___	say ___

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